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### File decay property

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#### Abstract

A computer-implemented method, computer system, and computer program product for deleting a file from a storage medium. A file that is marked for automatic deletion is identified. A deletion time at which the file is to be deleted from the storage medium is identified. A visual indicator associated with the file is displayed. The visual indicator changes over time to provide an indication as to a nearness of the deletion time.

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## Background/Summary

1. CROSS-REFERENCES TO RELATED APPLICATIONS (1) This application claims the benefit of priority under 35 U.S.C. § 120 as a continuation of U.S. patent application Ser. No. 16/896,892, filed Jun. 9, 2020, which is hereby incorporated by reference herein in its entirety.

## 2. FIELD

(1) The present disclosure relates generally to an improved computer system and, in particular, to managing electronic files in a computer system. More particularly, the present disclosure relates to establishing deletion times for electronic files, automatically deleting electronic files, and providing visual indicators for electronic files that are to be automatically deleted.

## 3. BACKGROUND

(2) The accumulation of unneeded files on a computer is a perennial problem. Many unneeded files stored on a computer take up storage space and may have an adverse effect on computer performance. Furthermore, the computer workspace view, that is, the computer desktop or window in which the files on the computer are listed, may become cluttered with many files, including files that are no longer needed along with files that are still being used, thereby making it more difficult for a user to locate a file that is needed. Removing unneeded files from a computer can be problematic, as a user may not be able to distinguish easily between needed files and unneeded files that may be deleted, based on the file names and other metadata in the computer workspace view alone.

(3) An example of a situation in which unneeded files may accumulate on a computer is in a collaborative working environment. In collaborative working environments, many different users may perform tasks on a project and therefore require access to electronic project files. For example, one user may perform work with or on a computer file or data while a second user performs other tasks with the same data or with the output or work product of the first user. Many times, users receive or download files to their individual computers to make working with or on the files more convenient and to avoid work interruptions when network access becomes unavailable or slow. Some of these downloaded files are temporary files that the user needs only for a certain period of time in order to perform their job function. For example, such temporary files may need to be stored for longer than a few hours or days, but need not be stored permanently. A user in a collaborative working environment may work on many different projects using many different temporary files over a period of time. The user easily may forget which downloaded files are no longer needed, leading to the accumulation of many files that take up space on the computer but that are no longer needed.

(4) Therefore, a system for more effective and efficient management of files on a computer is desirable.

## SUMMARY

(5) An illustrative embodiment provides a computer-implemented method for automatically deleting a file from a storage medium. A computer system identifies a file that is marked for automatic deletion. The computer system identifies a deletion time at which the file is to be automatically deleted from the storage medium. The computer system displays a visual indicator associated with the file. The visual indicator changes over time to provide an indication of nearness of the deletion time.

(6) Another illustrative embodiment provides a computer system for automatically deleting a file. The computer system includes a storage medium configured to store the file, a display system, and a processor unit. The processor unit is configured to identify the file that is marked for automatic deletion from the storage medium, identify a deletion time at which the file is to be automatically deleted from the storage medium, and display a visual indicator associated with the file on the display system. The visual indicator changes over time to provide an indication of nearness of the deletion time.

(7) Yet another illustrative embodiment provides a computer program product for automatically deleting a file. The computer program product comprises a computer readable storage media having program instructions embodied therewith. The program instructions are executable by a computer to cause the computer to perform a method of: identifying the file that is marked for

automatic deletion from a storage medium, identifying a deletion time at which the file is to be automatically deleted from the storage medium, and displaying a visual indicator associated with the file. The visual indicator changes over time to provide an indication of nearness of the deletion time.

(8) The features and functions can be achieved independently in various embodiments of the present disclosure or may be combined in yet other embodiments in which further details can be seen with reference to the following description and drawings.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The novel features believed characteristic of the illustrative embodiments are set forth in the appended claims. The illustrative embodiments, however, as well as a preferred mode of use, further objectives, and features thereof, will best be understood by reference to the following detailed description of an illustrative embodiment of the present disclosure when read in conjunction with the accompanying drawings, wherein:

(2) FIG. 1 is a block diagram of an information environment depicted in accordance with an illustrative embodiment;

(3) FIG. 2 is a block diagram of a file decay management system (FDMS) depicted in accordance with an illustrative embodiment;

(4) FIG. 3 is a flowchart of a method for file decay management depicted in accordance with an illustrative embodiment;

(5) FIG. 4 is a flowchart of another method for file decay management depicted in accordance with an illustrative embodiment;

(6) FIGS. 5-9 are examples of displays illustrating the changing appearance of a visual indicator for a file to be automatically deleted as the deletion time for the automatic deletion of the file approaches in accordance with an illustrative embodiment; and

(7) FIG. 10 is a block diagram of a data processing system in accordance with an illustrative embodiment.

### DETAILED DESCRIPTION

(8) The illustrative embodiments recognize and take into account one or more different considerations. For example, the illustrative embodiments recognize and take into account that users often receive computer readable files for use on their computer. These files may need to be stored temporarily, so that the user may use the files to perform various tasks. However, many such files do not need to be stored permanently, and should be deleted from the computer after use.

(9) The illustrative embodiments further recognize and take into account that users often forget to delete unneeded files when they are finished using them. Users often receive many such files, and many files that are no longer needed can accumulate on the computer. These files take up storage space on the computer that may be needed for other items. Furthermore, the file names or other visual indicators corresponding to these unneeded files may clutter the computer desktop view or other file view, making it more difficult for the user to locate a specific file when needed.

(10) The illustrative embodiments provide methods, systems, and computer program products for automatically deleting computer files after the expiration of a user configurable time period. The illustrative embodiments provide the user with a visual indicator of the nearness of the deletion time for a particular file. For example, in some illustrative embodiments, the file name or other visual indicator of the file is made more transparent as the nearness of the deletion time for the file approaches. The user is thereby reminded when a file is to be automatically deleted soon, giving the user an opportunity to modify the automatic deletion time for the file or remark the file to not be automatically deleted if the user has not finished using the file.

(11) Illustrative embodiments may be a system, a method, and/or a computer program product at any possible technical detail level of integration. The computer program product may include a computer readable storage medium (or media) having computer readable program instructions thereon for causing a processor to carry out aspects of the illustrative embodiments.

(12) The computer readable storage medium can be a tangible device that can retain and store instructions for use by an instruction execution device. The computer readable storage medium may be, for example, but is not limited to, an electronic storage device, a magnetic storage device, an optical storage device, an electromagnetic storage device, a semiconductor storage device, or any suitable combination of the foregoing. A non-exhaustive list of more specific examples of the computer readable storage medium includes the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a static random access memory (SRAM), a portable compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), a memory stick, a floppy disk, a mechanically encoded device such as punch-cards or raised structures in a groove having instructions recorded thereon, and any suitable combination of the foregoing. A computer readable storage medium, as used herein, is not to be construed as being transitory signals  custom character such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire.

(13) Computer readable program instructions described herein can be downloaded to respective computing/processing devices from a computer readable storage medium or to an external computer or external storage device via a network, for example, the Internet, a local area network, a wide area network and/or a wireless network. The network may comprise copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and/or edge servers. A network adapter card or network interface in each computing/processing device receives computer readable program instructions from the network and forwards the computer readable program instructions for storage in a computer readable storage medium within the respective computing/processing device.

(14) Computer readable program instructions for carrying out operations of the illustrative embodiments may be assembler instructions, instruction-set-architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, firmware instructions, state-setting data, configuration data for integrated circuitry, or either source code or object code written in any combination of one or more programming languages, including an object oriented programming language such as Smalltalk, C++, or the like, and procedural programming languages, such as the “C” programming language or similar programming languages. The computer readable program instructions may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer, or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider). In some embodiments, electronic circuitry including, for example, programmable logic circuitry, field-programmable gate arrays (FPGA), or programmable logic arrays (PLA) may execute the computer readable program instructions by utilizing state information of the computer readable program instructions to personalize the electronic circuitry, in order to perform aspects of the illustrative embodiments.

(15) Aspects of the illustrative embodiments are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems), and computer program products. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be

implemented by computer readable program instructions.

(16) These computer readable program instructions may be provided to a processor of a computer, or other programmable data processing apparatus, to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks. These computer readable program instructions may also be stored in a computer readable storage medium that can direct a computer, a programmable data processing apparatus, and/or other devices to function in a particular manner, such that the computer readable storage medium having instructions stored therein comprises an article of manufacture including instructions which implement aspects of the function/act specified in the flowchart and/or block diagram block or blocks.

(17) The computer readable program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational steps to be performed on the computer, other programmable apparatus, or other device to produce a computer implemented process, such that the instructions which execute on the computer, other programmable apparatus, or other device implement the functions/acts specified in the flowchart and/or block diagram block or blocks.

(18) The flowcharts and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various illustrative embodiments. In this regard, each block in the flowcharts or block diagrams may represent a module, segment, or portion of instructions, which comprises one or more executable instructions for implementing the specified logical function(s). In some alternative implementations, the functions noted in the blocks may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be accomplished as one step, executed concurrently, substantially concurrently, in a partially or wholly temporally overlapping manner, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustrations, and combinations of blocks in the block diagrams and/or flowchart illustrations, can be implemented by special purpose hardware-based systems or combinations of special purpose hardware and computer instructions that perform the specified functions or acts.

(19) With reference now to the figures and, in particular, with reference to FIG. 1, a pictorial representation of a network of data processing systems is depicted in which illustrative embodiments may be implemented. Network data processing system **100** is a network of computers in which the illustrative embodiments may be implemented. Network data processing system **100** contains network **102**, which is the medium used to provide communications links between various devices and computers connected together within network data processing system **100**. Network **102** may include connections, such as wire, wireless communication links, or fiber optic cables.

(20) In the depicted example, server computer **104** and server computer **106** connect to network **102** along with storage unit **108**. In addition, client devices **110** connect to network **102**. Client devices **110** can be, for example, computers, workstations, or network computers. As depicted, client devices **110** include client computer **112**, client computer **114**, and client computer **116**. Further, client devices **110** can also include other types of devices, such mobile phone **118**, tablet computer **120**, smart speaker **122**, and smart glasses **124**. In the depicted example, server computer **104** provides information, such as boot files, operating system images, and applications to client devices **110**. In this illustrative example, server computer **104**, server computer **106**, storage unit **108**, and client devices **110** are network devices that connect to network **102** in which network **102** is the communications media for these network devices.

(21) Client devices **110** are clients to server computer **104** in this example. Network data processing system **100** may include additional server computers, client computers, and other devices not shown. Client devices **110** connect to network **102** utilizing at least one of wired,

optical fiber, or wireless connections.

(22) Program code located in network data processing system **100** can be stored on a computer-recordable storage medium and downloaded to a data processing system or other device for use. For example, program code can be stored on a computer-recordable storage medium on server computer **104** and downloaded to client devices **110** over network **102** for use on client devices **110**.

(23) In the depicted example, network data processing system **100** is the Internet with network **102** representing a worldwide collection of networks and gateways that use the Transmission Control Protocol/Internet Protocol (TCP/IP) suite of protocols to communicate with one another. At the heart of the Internet is a backbone of high-speed data communication lines between major nodes or host computers consisting of thousands of commercial, governmental, educational, and other computer systems that route data and messages. Of course, network data processing system **100** also may be implemented using a number of different types of networks. For example, network **102** can be comprised of at least one of the Internet, an intranet, a local area network (LAN), a metropolitan area network (MAN), or a wide area network (WAN). Network **102** may be comprised of the Internet-of-Things (IoT). FIG. **1** is intended as an example, and not as an architectural limitation for the different illustrative embodiments.

(24) As used herein, “a number of” when used with reference to items, means one or more items. For example, “a number of different types of networks” is one or more different types of networks.

(25) Further, the phrase “at least one of,” when used with a list of items, means different combinations of one or more of the listed items can be used, and only one of each item in the list may be needed. In other words, “at least one of” means any combination of items and number of items may be used from the list, but not all of the items in the list are required. The item can be a particular object, a thing, or a category.

(26) For example, without limitation, “at least one of item A, item B, or item C” may include item A, item A and item B, or item B. This example also may include item A, item B, and item C or item B and item C. Of course, any combinations of these items can be present. In some illustrative examples, “at least one of” can be, for example, without limitation, two of item A; one of item B; and ten of item C; four of item B and seven of item C; or other suitable combinations.

(27) Turning now to FIG. **2**, a block diagram of a file decay management system (FDMS) **200** is depicted in accordance with an illustrative embodiment. FDMS **200** is implemented in computer system **201**. Computer system **201** is a physical hardware system that includes one or more data processing systems. When more than one data processing system is present, those data processing systems may be in communication with each other using a communications medium. The communications medium may be a network. The data processing systems may be selected from at least one of a computer, a server computer, a workstation, a tablet computer, a laptop computer, smart glasses, a mobile phone, or some other suitable data processing system. FDMS **200** may be implemented on, for example, any of client devices **110** or server computers **104**, **106** depicted in FIG. **1**. In an embodiment, FDMS **200** may be implemented in a distributed computing environment in which some of the components of FDMS **200** are executed on server computer **104** while other components of FDMS **200** are executed on one of client devices **110**.

(28) FDMS **200** provides a system for allowing a user **202** to manage the deletion of a file **203** from a storage medium **204**. The file **203** may include any electronic content that is intended to be used in either an electronic form or as printed output. For example, without limitation, the file **203** may be referred to as an electronic file, a computer file, an electronic document, or using other appropriate terms.

(29) The storage medium **204** may include any appropriate medium or device on which the file **203** may be stored and from which the file **203** may be deleted. For example, without limitation, storage medium **204** may include storage devices **1016** or computer readable media **1020** as described in more detail below with reference to FIG. **10**.

(30) In accordance with an illustrative embodiment, the FDMS **200** is configured to allow the user **202** to delete the file **203** from the storage medium **204** immediately or to designate the file **203** for automatic deletion from the storage medium **204** after a specified time period. The FDMS **200** also provides a visual indicator of the nearness of the automatic deletion time for the file **203** and enables the user **202** to modify the automatic deletion time for the file **203** or to mark the file **203** to not be automatically deleted, if the user **202** changes their mind about deleting the file **203** before the automatic deletion time. The functionality provided by the FDMS **200** enables the user **202** to reduce the number of unneeded files on their computer in an effective and efficient manner, so that the user **202** is not overwhelmed with too much clutter on the display when looking for a particular file, and so that the storage for the computer does not become loaded with files that are no longer needed, and thus more storage space is available for files that are needed.

(31) As depicted, FDMS **200** includes a user interface **205**, a file decay attributes storage **216**, a deletion engine **232**, a display manager **234**, and a calendar **236**.

(32) In this illustrative example, the user interface **205** can be in the form of a graphical user interface. As depicted, the user interface **205** can be implemented in human machine interface (HMI) **206**. HMI **206** is a hardware system that includes display system **207** and input system **208**. Display system **207** is a physical hardware system and includes one or more display devices on which user interface **205** can be displayed. The display devices can include at least one of a light emitting diode (LED) display, a liquid crystal display (LCD), an organic light emitting diode (OLED) display, a computer monitor, a projector, a flat panel display, a heads-up display (HUD), or some other suitable device that can output information for the visual presentation of information.

(33) The user **202** is a person that can interact with the user interface **205** through user input generated by the input system **208**. The input system **208** is a physical hardware system and can be selected from at least one of a mouse, a keyboard, a trackball, a touchscreen, a stylus, a motion sensing input device, a gesture detection device, a cyber glove, or some other suitable type of input device.

(34) In an illustrative embodiment, the user interface **205** may be accessed and engaged by, for example, the user **202** using the input system **208** to right click on or otherwise select a visual indicator displayed on the display system **207** for the file **203** which the user **202** wishes to have marked for automatic deletion or unmarked to not be automatically deleted. In this illustrative example, the user interface **205** includes a marking for automatic deletion interface **209** and an unmarking for automatic deletion interface **210**.

(35) The marking for automatic deletion interface **209** is configured to allow the user **202** to designate the file **203** for automatic deletion. The marking for automatic deletion interface **209** also is configured to allow the user **202** to specify details for the automatic deletion of the file **203**.

(36) In an illustrative embodiment, the marking for automatic deletion interface **209** includes a file identity entry **211**, a deletion time entry **212**, and a display type entry **214**. The file identity entry **211** is configured to allow the user **202** to identify the file **203** to be automatically deleted in any appropriate manner.

(37) The deletion time entry **212** is configured to allow the user **202** to identify a deletion time for the automatic deletion of the file **203** from the storage medium **204** in any appropriate manner. For example, the deletion time entry **212** may be configured to allow the operator **202** to identify the deletion time by indicating a specific date for automatically deleting the file **203**. Alternatively, the deletion time entry **212** may be configured to allow the operator **202** to identify the deletion time by entering a length of time from the current time, such as, for example, without limitation, a number of days from the current date, for automatically deleting the file **203**.

(38) The display type entry **214** may be configured to allow the user **202** to specify the type of visual indicator that will be used to indicate that the file **203** is marked for automatic deletion. The display type entry **214** also may be configured to allow the user **202** to specify how the visual indicator will change over time to indicate the nearness of the deletion time.



(39) The unmarking for automatic deletion interface **210** is configured to allow the user **202** to mark the file **203** to not be automatically deleted. The user **203** may use the unmarking for automatic deletion interface **210** to prevent the file **203** from being automatically deleted at any time before the file **203** is automatically deleted from the storage medium **204**.

(40) The file decay attributes storage **216** includes file decay attributes **222** for each file to be automatically deleted **220** in a number of files to be automatically deleted **218**. The number of files to be automatically deleted **218** and the file decay attributes **222** for each file to be automatically deleted **220** in the number of files to be automatically deleted **218** may be identified by the user **202** using the marking for automatic deletion interface **209**.

(41) The file decay attributes **222** for each file to be automatically deleted **220** may include a file identifier **224**, a deletion time **226**, a display type **228**, and a visual indicator for current time **230**. The file identifier **224** may identify the file to be automatically deleted **220** in any appropriate manner. For example, without limitation, the file identifier **224** may identify the location in the storage medium **204** of the file to be automatically deleted.

(42) The deletion time **226** may identify the time for automatic deletion of the file to be automatically deleted **220** in any appropriate manner. For example, without limitation, deletion time **226** may be a deletion date **227** for automatic deletion of the file to be automatically deleted **220**.

(43) The display type **228** indicates how a visual indicator **229** for the file to be automatically deleted **220** is displayed to the user **202** on the display system **207** to indicate that the file is marked for automatic deletion. For example, the visual indicator **229** may include alphanumeric text, such as the name of the file to be automatically deleted **220**, a graphical icon, another appropriate visual indicator, or a combination of alphanumeric text and graphics. The display type **228** also may indicate how the visual indicator **229** for the file to be automatically deleted **220** changes over time as the deletion time **226** for automatic deletion of the file to be automatically deleted **220** nears. The changing visual indicator **229** for the file to be automatically deleted **220** reminds the user **202** that the file is marked for automatic deletion and indicates how soon the file will be automatically deleted. The user **202** then has the opportunity to unmark the file for automatic deletion, if desired, so the file will not be automatically deleted.

(44) For example, without limitation, the display type **228** may indicate that the visual indicator **229** for the file to be automatically deleted **220** becomes more transparent as the deletion time **226** for automatic deletion of the file to be automatically deleted **220** approaches. The transparency of the visual indicator **229** for the file to be automatically deleted **220** may be defined relative to the typical visual indicator for a file that is not marked for automatic deletion. For example, a visual indicator **229** for the file to be automatically deleted **220** that is 0% transparent is completely visible to the same extent as other visual indicators for files that are not marked for automatic deletion. A visual indicator **229** for the file to be automatically deleted **220** that is 80% transparent is almost invisible, although still visible. For example, without limitation, the percentage of transparency of the visual indicator **229** may indicate the percentage of pixels of a visual indicator **229** that is 0% transparent that are replaced by the corresponding pixels of the background immediately behind the visual indicator **229**. The desired level of transparency of the visual indicator **229** may be implemented in any other appropriate manner.

(45) The display type **228** may define any appropriate relationship between the transparency of the visual indicator **229** for the file to be automatically deleted **220** and the amount of time until the deletion time **226** for automatic deletion of the file to be automatically deleted **220**. For example, the visual indicator **229** for the file to be automatically deleted **220** may be 0% transparent until a decay start time is reached. For example, without limitation, the decay start time may be a fixed number of days before the deletion time **226** for the file to be automatically deleted **220**, a number of days before the deletion time **226** that is selectable by the user **202**, or any other appropriate amount of time before the deletion time **226**. Starting at the decay start time, the transparency of

the visual indicator **229** for the file to be automatically deleted **220** may be increased from 0% transparency at an appropriate rate to indicate that the deletion time **226** for automatic deletion of the file to be automatically deleted **220** is approaching. For example, starting at the decay start time, the transparency of the visual indicator **229** may be increased linearly, exponentially, or in any other appropriate manner.

(46) The transparency of the visual indicator **229** may be described in terms of opacity. Opacity is the opposite of transparency. For example, visual indicator **229** with 0% transparency may be described as 100% opaque. Visual indicator **229** with 80% transparency may be described as 20% opaque. Therefore, in accordance with an illustrative embodiment, the display type **228** may indicate that the visual indicator **229** for the file to be automatically deleted **220** becomes less opaque as the deletion time **226** for automatic deletion of the file to be automatically deleted **220** approaches.

(47) As another example, the display type **228** may indicate that the color of the visual indicator **229** for the file to be automatically deleted **220** changes as the deletion time **226** for automatic deletion of the file to be automatically deleted **220** approaches. For example, without limitation, the display type **228** may indicate that the color of the visual indicator **229** for the file to be automatically deleted **220** changes over time, from, for example, green, to yellow, to red, as the deletion time **226** for automatic deletion of the file to be automatically deleted **220** becomes closer.

(48) The visual indicator for current time **230** indicates how the visual indicator **229** for the file to be automatically deleted **220** is currently to be displayed on the display system **207**. The visual indicator for current time **230** is determined based on the current time, how near the current time is to the deletion time **226** for automatically deleting the file to be automatically deleted **220**, and the display type **228** indicating how the proximity of the current time to the deletion time **226** is to be indicated in the visual indicator **229** for the file to be automatically deleted **220**.

(49) The deletion engine **232** is configured to analyze the file decay attributes **222** for the number of files to be automatically deleted **218** identified in the file decay attributes storage **216**. The deletion engine **232** is configured to compare the deletion time **226** for each file to be automatically deleted **220** to the current time **237** obtained from a calendar **236** to determine the nearness of the deletion time **226** for each file to be automatically deleted **220**. The deletion engine **232** is configured to generate or modify the visual indicator for current time **230** for each file to be automatically deleted **220** according to the determined nearness of the deletion time **226** for the file to be automatically deleted **220** and in the manner defined by the display type **228** for the file to be automatically deleted **220**.

(50) The deletion engine **232** also is configured to delete automatically the file to be automatically deleted **220** from the storage medium **204** in response to the deletion time **226** for the file to be automatically deleted **220** being reached. For example, the deletion time **226** for the file to be automatically deleted **220** may be determined to be reached when the current time **237** is the same as or after the deletion time **226** for the file to be automatically deleted **220**. The deletion engine **232** also may remove the file decay attributes **222** for the file to be automatically deleted **220** from the file decay attributes storage **216** when the file to be automatically deleted **220** is deleted from the storage medium **204**.

(51) The display manager **234** is configured to control the display of the visual indicator **229** for each file to be automatically deleted **220**. Whenever the visual indicator **229** for a file to be automatically deleted **220** is requested or required to be displayed in the display system **207**, the display manager **234** ensures that the visual indicator **229** displayed for the file to be automatically deleted **220** is the visual indicator for current time **230**, which indicates how close the file to be automatically deleted **220** is to being deleted.

(52) The illustration of file decay management system **200** in FIG. 2 is not meant to imply physical or architectural limitations to the manner in which illustrative embodiments may be implemented. Other components, in addition to or in place of the ones illustrated, may be used. Some components

may be optional. Also, the blocks are presented to illustrate some functional components. One or more of these blocks may be combined, divided, or combined and divided into different blocks when implemented in an illustrative embodiment.

(53) Turning now to FIG. 3, a flowchart of a method **300** for file decay management is depicted in accordance with an illustrative embodiment. The method **300** may be implemented, for example, in file decay management system **200** in FIG. 2.

(54) The method **300** begins by determining if a file is marked for automatic deletion (step **302**). If the file is not marked for automatic deletion, a normal visual indicator associated with the file may be displayed (step **303**) and the file is not automatically deleted (step **304**), with the method **300** ending thereafter.

(55) In response to a determination at step **302** that the file is marked for automatic deletion, a deletion time at which the file is to be deleted automatically from a storage medium is determined (step **305**). The deletion time may be user selectable. Alternatively, the deletion time may be set automatically to a predefined value in response to the user designating the file to be deleted automatically.

(56) The method **300** continues by displaying a visual indicator associated with the file that changes over time to indicate the nearness of the deletion time for the file (step **306**). The visual indicator displayed in step **306** may take a number of different forms, depending on the particular embodiment. For example, the visual indicator may include text, such as a file name for the file, a graphical icon, or both text and graphics as used to identify files in a computer system. The manner in which the visual indicator for the file changes over time as the deletion time for automatic deletion of the file becomes closer also may take a number of different forms. For example, the visual indicator for the file may become more transparent over time as the deletion time gets closer. In another embodiment, the color of the visual indicator may change over time as the deletion time gets closer. For example, the color of the visual indicator for the file may change from green to yellow as the deletion time for the automatic deletion of the file becomes closer. The color of the visual indicator may then change to red when the deletion time for the automatic deletion of the file is very near, for example, without limitation, within the next day. In another embodiment, the shape of the visual indicator may change over time as the deletion time becomes closer. For example, the visual indicator may change from one particular graphical icon to a different graphical icon over time as the deletion time for the automatic deletion of the file becomes closer. The way in which the visual indicator changes over time to indicate the nearness of the deletion time for the automatic deletion of the file may be user selectable.

(57) The method **300** continues by determining whether the user has unmarked the file for automatic deletion (step **308**). The file can be unmarked for automatic deletion in step **308** in response to receiving a user input. If the user has unmarked the file for automatic deletion at step **308**, the method **300** proceeds to display a normal visual indicator for the file at step **303** and to not automatically delete the file at step **304**, with the method **300** ending thereafter.

(58) In response to a determination at step **308** that the user has not unmarked the file for automatic deletion, the method **300** proceeds to determine whether the current time is the same as or after the deletion time for the automatic deletion of the file (step **312**). In response to a determination at step **312** that the deletion time has been reached, the file is automatically deleted from the storage medium (step **314**), after which the method **300** may end. In response to a determination at operation **312** that the deletion time has not yet been reached, the method **300** may return to step **306**, such that the visual indicator for the file changes to indicate the nearness of the deletion time for the automatic deletion of the file until the file is either automatically deleted or unmarked for automatic deletion.

(59) Turning now to FIG. 4, a flowchart of another method **400** for file decay management is depicted in accordance with an illustrative embodiment. The method **400** is an example of one possible implementation of portions of the method **300** in FIG. 3. The method **400** may be

implemented, for example, in file decay management system **200** in FIG. 2.

(60) The method **400** begins by identifying a file in a storage medium (step **402**). A user then may mark the file for automatic deletion, set a deletion time for the automatic deletion of the file, and save the deletion time and other file decay attributes for the file to be automatically deleted (step **404**). The deletion time for the automatic deletion of the file then may be compared to the current time (step **406**) to determine whether the deletion time for the automatic deletion of the file has been reached (step **408**). In response to a determination at step **408** that the deletion time has been reached, the file is automatically deleted from the storage medium and a visual indicator associated with the file is no longer displayed (step **410**), with the method **400** ending thereafter.

(61) In response to a determination at step **408** that the deletion time for the automatic deletion of the file has not been reached, the method **400** may proceed to calculate the number of days from the current time to the deletion time (step **412**). The method **400** then may determine whether the number of days to the deletion time is three days (step **414**). If the number of days to the deletion time is determined to be three days at step **414**, the method **400** may set the transparency of the visual indicator associated with the file to 20% transparency (step **416**), with the method **400** returning to step **406** thereafter. If the number of days to the deletion time is not determined to be three days at step **414**, the method **400** may determine whether the number of days to the deletion time is two days (step **418**). If the number of days to the deletion time is determined to be two days at step **418**, the method **400** may set the transparency of the visual indicator associated with the file to 40% transparency (step **420**), with the method **400** returning to step **406** thereafter. If the number of days to the deletion time is not determined to be two days at step **418**, the method **400** may determine whether the number of days to the deletion time is one day (step **422**). If the number of days to the deletion time is determined to be one day at step **422**, the method **400** may set the transparency of the visual indicator associated with the file to 60% transparency (step **424**), with the method **400** returning to step **406** thereafter. If the number of days to the deletion time is not determined to be one day at step **422**, the method **400** may set the transparency of the visual indicator associated with the file to 0% transparency (step **426**), with the method **400** returning to step **406** thereafter.

(62) Turning now to FIGS. 5-9, displays **500**, **600**, **700**, **800**, and **900** are examples of displays illustrating the changing appearance of a visual indicator for a file to be automatically deleted as the deletion time for the automatic deletion of the file approaches in accordance with an illustrative embodiment. Displays **500**, **600**, **700**, **800**, and **900** are examples of displays that can be displayed in human machine interface **206**, in particular in user interface **205** on display system **207**, in file decay management system **200** in FIG. 2.

(63) In display **500** in FIG. 5, a visual indicator **502** for a file with the file name “ADP\_Health\_Compliance\_ACA\_IRS\_Reporting-unlocked-do not distribute.pdf” is displayed in a normal manner, in the same manner as the visual indicators **504** for other files. This indicates that the file is not designated for automatic deletion in accordance with an illustrative embodiment or that the deletion time for the automatic deletion of the file is relatively far in the future. In display **600** in FIG. 6, the deletion time for the automatic deletion of the file indicated by the visual indicator **502** is three days away. In this example, the visual indicator **502** for the file to be deleted is displayed in display **600** at 20% transparency. In display **700** in FIG. 7, the deletion time for the automatic deletion of the file indicated by the visual indicator **502** is two days away. In this example, the visual indicator **502** for the file to be deleted is displayed in display **700** at 40% transparency. In display **800** in FIG. 8, the deletion time for the automatic deletion of the file indicated by the visual indicator **502** is one day away. In this example, the visual indicator **502** for the file to be deleted is displayed in display **800** at 60% transparency. In display **900** in FIG. 9, the deletion time for the automatic deletion of the file indicated by the visual indicator **502** in displays **500**, **600**, **700**, and **800** has been reached. In this example, the file has been automatically deleted and no visual indicator for the file is displayed in display **900**.

(64) Turning now to FIG. 10, an illustration of a block diagram of a data processing system **1000** is depicted in accordance with an illustrative embodiment. Data processing system **1000** is an example of computer system **201** on which file decay management system **200** in FIG. 2 may be implemented.

(65) In this illustrative example, data processing system **1000** includes communications framework **1002**, which provides communications between processor unit **1004**, memory **1006**, persistent storage **1008**, communications unit **1010**, input/output unit **1012**, and display **1014**. In this example, communications framework **1002** may take the form of a bus system.

(66) Processor unit **1004** serves to execute instructions for software that may be loaded into memory **1006**. Processor unit **1004** may be a number of processors, a multi-processor core, or some other type of processor, depending on the particular implementation. In an embodiment, processor unit **1004** comprises one or more conventional general purpose central processing units (CPUs). In an alternate embodiment, processor unit **1004** comprises one or more graphical processing units (GPUs).

(67) Memory **1006** and persistent storage **1008** are examples of storage devices **1016**. A storage device is any piece of hardware that is capable of storing information, such as, for example, without limitation, at least one of data, program code in functional form, or other suitable information either on a temporary basis, a permanent basis, or both on a temporary basis and a permanent basis. Storage devices **1016** may also be referred to as computer-readable storage devices in these illustrative examples. Memory **1006**, in these examples, may be, for example, a random access memory or any other suitable volatile or non-volatile storage device. Persistent storage **1008** may take various forms, depending on the particular implementation.

(68) For example, persistent storage **1008** may contain one or more components or devices. For example, persistent storage **1008** may be a hard drive, a flash memory, a rewritable optical disk, a rewritable magnetic tape, or some combination of the above. The media used by persistent storage **1008** also may be removable. For example, a removable hard drive may be used for persistent storage **1008**. Communications unit **1010**, in these illustrative examples, provides for communications with other data processing systems or devices. In these illustrative examples, communications unit **1010** is a network interface card.

(69) Input/output unit **1012** allows for input and output of data with other devices that may be connected to data processing system **1000**. For example, input/output unit **1012** may provide a connection for user input through at least one of a keyboard, a mouse, or some other suitable input device. Further, input/output unit **1012** may send output to a printer. Display **1014** provides a mechanism to display information to a user.

(70) Instructions for at least one of the operating system, applications, or programs may be located in storage devices **1016**, which are in communication with processor unit **1004** through communications framework **1002**. The processes of the different embodiments may be performed by processor unit **1004** using computer-implemented instructions, which may be located in a memory, such as memory **1006**.

(71) These instructions are referred to as program code, computer-usable program code, or computer-readable program code that may be read and executed by a processor in processor unit **1004**. The program code in the different embodiments may be embodied on different physical or computer-readable storage media, such as memory **1006** or persistent storage **1008**.

(72) Program code **1018** is located in a functional form on computer-readable media **1020** that is selectively removable and may be loaded onto or transferred to data processing system **1000** for execution by processor unit **1004**. Program code **1018** and computer-readable media **1020** form computer program product **1022** in these illustrative examples. In one example, computer-readable media **1020** may be computer-readable storage media **1024** or computer-readable signal media **1026**.

(73) In these illustrative examples, computer-readable storage media **1024** is a physical or tangible

storage device used to store program code **1018** rather than a medium that propagates or transmits program code **1018**. Alternatively, program code **1018** may be transferred to data processing system **1000** using computer-readable signal media **1026**.

(74) Computer-readable signal media **1026** may be, for example, a propagated data signal containing program code **1018**. For example, computer-readable signal media **1026** may be at least one of an electromagnetic signal, an optical signal, or any other suitable type of signal. These signals may be transmitted over at least one of communications links, such as wireless communications links, optical fiber cable, coaxial cable, a wire, or any other suitable type of communications link.

(75) Further, as used herein, “computer-readable media **1020**” can be singular or plural. For example, program code **1018** can be located in computer-readable media **1020** in the form of a single storage device or system. In another example, program code **1018** can be located in computer-readable media **1020** that is distributed in multiple data processing systems. In other words, some instructions in program code **1018** can be located in one data processing system while other instructions in program code **1018** can be located in another data processing system. For example, a portion of program code **1018** can be located in computer-readable media **1020** in a server computer while another portion of program code **1018** can be located in computer-readable media **1020** located in a set of client computers.

(76) The different components illustrated for data processing system **1000** are not meant to provide architectural limitations to the manner in which different embodiments may be implemented. The different illustrative embodiments may be implemented in a data processing system including components in addition to or in place of those illustrated for data processing system **1000**. Other components shown in FIG. **10** can be varied from the illustrative examples shown. The different embodiments may be implemented using any hardware device or system capable of running program code **1018**.

(77) The flowcharts and block diagrams in the different depicted embodiments illustrate the architecture, functionality, and operation of some possible implementations of apparatuses and methods in an illustrative embodiment. In this regard, each block in the flowcharts or block diagrams may represent at least one of a module, a segment, a function, or a portion of an operation or step. For example, one or more of the blocks may be implemented as program code.

(78) In some alternative implementations of an illustrative embodiment, the function or functions noted in the blocks may occur out of the order noted in the figures. For example, in some cases, two blocks shown in succession may be performed substantially concurrently, or the blocks may sometimes be performed in the reverse order, depending upon the functionality involved. Also, other blocks may be added to the illustrated blocks in a flowchart or block diagram.

(79) The description of the different illustrative embodiments has been presented for purposes of illustration and description and is not intended to be exhaustive or to limit the embodiments in the form disclosed. The different illustrative examples describe components that perform actions or operations. In an illustrative embodiment, a component may be configured to perform the action or operation described. For example, the component may have a configuration or design for a structure that provides the component an ability to perform the action or operation that is described in the illustrative examples as being performed by the component. Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different illustrative embodiments may provide different features as compared to other desirable embodiments. The embodiment or embodiments selected are chosen and described in order to best explain the principles of the embodiments, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

## Claims

1. A system, comprising: one or more processors, coupled with memory, to: identify, responsive to an instruction, that an electronic file stored on a storage medium is marked for deletion from the storage medium at a deletion time and date, the instruction indicating the deletion time and date for the electronic file and a type of indicator to use for the electronic file; determine a level of transparency based on a difference between a current time and the deletion time and date; configure, responsive to the instruction and in accordance with the type of indicator for the electronic file, an indicator with the level of transparency that increases as a countdown relative to the deletion time and date decreases; and provide, for display via a display device at a time prior to the deletion time and date, the indicator for the electronic file configured with the level of transparency determined based on the countdown at the current time relative to the deletion time and date.
2. The system of claim 1, wherein the instruction marks the electronic file for deletion from the storage medium at the deletion time and date.
3. The system of claim 1, wherein the indicator has: a first level of transparency three days before the deletion time and date, a second level of transparency two days before the deletion time and date that is greater than the first level of transparency, and a third level of transparency one day before the deletion time and date that is greater than the second level of transparency.
4. The system of claim 3, wherein the second level of transparency is at least twice the first level of transparency.
5. The system of claim 3, wherein the third level of transparency is at least twice the second level of transparency.
6. The system of claim 1, comprising: the one or more processors to automatically delete the electronic file at the deletion time and date.
7. The system of claim 1, comprising the one or more processors to: receive, via a user interface at a time prior to the deletion time, an instruction to unmark the electronic file for deletion at the deletion time and date; and provide, for display via the display device responsive to the instruction, the indicator without any level of transparency.
8. The system of claim 1, wherein a color of the indicator changes based on the countdown relative to the deletion time and date.
9. The system of claim 1, comprising the one or more processors to: retrieve, from the memory, file decay attributes established for the electronic file; and determine the level of transparency for the electronic file based on the file decay attributes.
10. The system of claim 1, comprising the one or more processors to: download, for storage in the storage medium, the electronic file from a collaborative working environment used by a plurality of users; and establish the deletion time and date for the electronic file responsive to download.
11. A method, comprising: identifying, by one or more processors coupled with memory, responsive to an instruction, that an electronic file stored on a storage medium is marked for deletion from the storage medium at a deletion time and date, the instruction indicating the deletion time and date for the electronic file and a type of indicator for the electronic file; determining, by the one or more processors, a level of transparency based on a difference between a current time and the deletion time and date; configuring, by the one or more processors, responsive to the instruction and in accordance with the type of indicator for the electronic file, an indicator with the level of transparency that increases as a countdown relative to the deletion time and date decreases; and providing, by the one or more processors for display via a display device at a time prior to the deletion time and date, the indicator for the electronic file configured with the level of transparency determined based on the countdown at the current time relative to the deletion time and date.
12. The method of claim 11, wherein the instruction marks the electronic file for deletion at the

deletion time and date.

13. The method of claim 11, wherein the indicator has: a first level of transparency three days before the deletion time and date, a second level of transparency two days before the deletion time and date that is greater than the first level of transparency, and a third level of transparency one day before the deletion time and date that is greater than the second level of transparency.

14. The method of claim 13, wherein the second level of transparency is at least twice the first level of transparency, and the third level of transparency is at least twice the second level of transparency.

15. The method of claim 11, comprising: automatically deleting, by the one or more processors, the electronic file at the deletion time and date.

16. The method of claim 11, comprising: receiving, by the one or more processors, via a user interface at a time prior to the deletion time, an instruction to unmark the electronic file for deletion at the deletion time and date; and providing, by the one or more processors, for display via the display device responsive to the instruction, the indicator without any level of transparency.

17. The method of claim 11, comprising: retrieving, by the one or more processors from the memory, file decay attributes established for the electronic file; and determining, by the one or more processors, the level of transparency for the electronic file based on the file decay attributes.

18. The method of claim 11, comprising: downloading, by the one or more processors for storage in the storage medium, the electronic file from a collaborative working environment used by a plurality of users; and establishing, by the one or more processors, the deletion time and date for the electronic file responsive to download.

19. A non-transitory computer-readable medium comprising processor executable instructions that, when executed by one or more processors, cause the one or more processors to: identify, responsive to an instruction, that an electronic file stored on a storage medium is marked for deletion from the storage medium at a deletion time and date, the instruction indicating the deletion time and date for the electronic file and a type of indicator for the electronic file; determine a level of transparency based on a difference between a current time and the deletion time and date; configure, responsive to the instruction and in accordance with the type of indicator for the electronic file, an indicator with the level of transparency that increases as a countdown relative to the deletion time and date decreases; and provide, for display via a display device at a time prior to the deletion time and date, the indicator for the electronic file configured with the level of transparency determined based on the countdown at the current time relative to the deletion time and date.

20. The non-transitory computer-readable medium of claim 19, wherein the instruction marks the electronic file for deletion at the deletion time and date.

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